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"SQ's of Inorganic Chemistry"

2019

1. Describe metallic bond on basis of Band theory.

Ans: Metallic Bond: Metals are those materials in which plenty of electrons are available for electric conduction. In terms of energy bands the conduction band of conductor is partly filled and contains a very large no. of electrons. These electrons can be promoted to the nearby vacant levels by electric or thermal energy. This explains conductivity in metals.

2. What is effect of impurity on conductivity of metals.

A: Effect of impurity: In most metals, the existence of impurities restricts the flow of electrons. Compared to pure metals, then, elements which are added as alloying agents could be considered "impurities". So alloys tend to offer less electrical conductivity than pure metal.

Q: Give postulates of Werner's theory?

A: Werner's Theory: The important postulates of Werner's theory are:-

- The central metal or the metal atoms in coordination compounds show two types of valency.
- The primary valency relates to the oxidation state and secondary valency relates to coordination no.

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• The no. of secondary valencies is fixed for every metal atom.

4: What are Magic no.? Give examples.

Ans: Magic Numbers:-

magic nos. are $\{2, 8, 20, 28, 50, 82, 126\}$ ^{the conventional}
These numbers were found in case of protons by comparing the no. of stable isotopes for different isotopes for diff. proton numbers.

For neutrons the magic no. were found by comparing no. of stable nuclides with the same neutron no.

5: Write a short note on Semiconductors

Ans: Semiconductors: Semiconductors are materials which have a conductivity b/w conductors (generally metals) and non-conductors. Semiconductors can be compounds such as Gallium arsenide or pure elements, such as Germanium or Silicon.

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Q6: what are outer orbital complexes?
Give examples.

Ans: Outer orbital complexes:-

Outer orbital complexes are coordination compounds composed of a central metal atom having hybridization of ~~outer~~ atomic orbitals including s, p and d orbitals from the outermost shell.

e.g., hexaquo nickel (II) ion, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
hexamine zinc (II) ion, $[\text{Zn}(\text{NH}_3)_6]^{2+}$

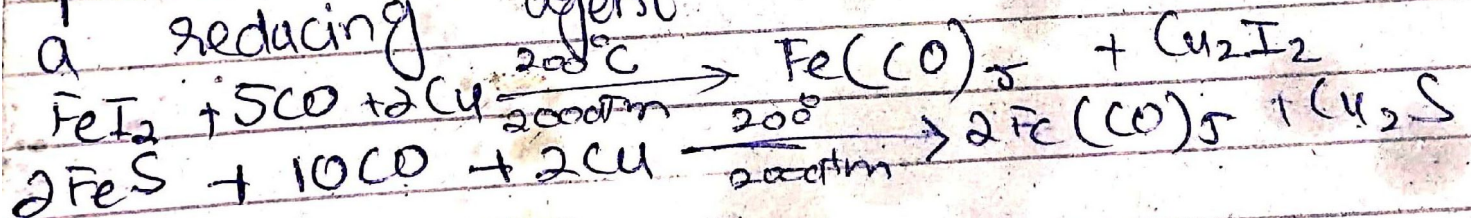
Q7: Give three methods of preparation of $\text{Fe}(\text{CO})_5$?

Ans: Methods of Preparation:-

$\text{Fe}(\text{CO})_5$ is prepared:-

(i) by the direct combination of CO & iron metal
$$\text{Fe} + 5\text{CO} \xrightarrow[2000\text{ atm}]{200^\circ\text{C}} \text{Fe}(\text{CO})_5$$

(ii) by carbonylating FeI_2 or FeS with CO in presence of Cu-metal which acts as a reducing agent.



Q8: What are limitations of CFT?

A: Limitations of CFT:-

⇒ The assumption that the interaction b/w metal-ligand is purely electrostatic cannot be said to be very realistic.

⇒ This theory only takes d-orbitals of central atom in account.

⇒ The theory fails to explain the behaviour of certain metals which cause large splitting while others shows small splitting.

“Past Papers - (2018)”

Q1: What are intrinsic semi-conductors? Explain with the help of $\mu(E)$ curves.

Ans: Intrinsic Semi-conductors:-

An intrinsic

(pure) semiconductor also called undoped or i-type semiconductors is a pure semi-conductor without any significant dopant species present. The no. of

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charge carriers is therefore determined by properties of the material itself instead of the amount of impurities

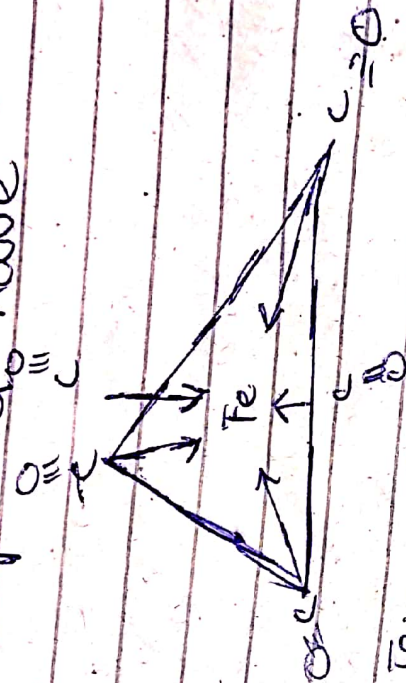
2: show the hybridization of $Fe(CO)_5$ with structure?

Ans: Structure & hybridization:-

Electron diffraction and infrared studies of this compound have shown that

$Fe(CO)_5$ molecule

has trigonal bipyramidal shape.



dsp^2 hybridization of

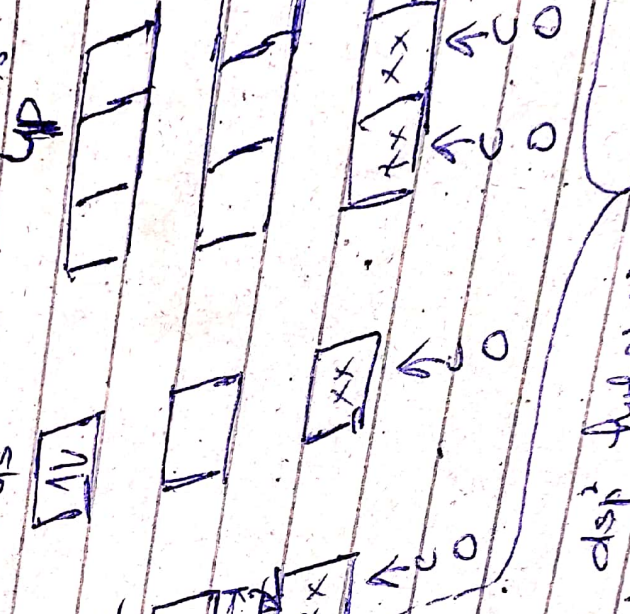
Fe atom:-

Trigonal bipyramidal structure of $Fe(CO)_5$:-

Free atom (3d-4s-4p) 3d

Fe atom in $Fe(CO)_5$ (3d-4s-4p)

$Fe(CO)_5$ molecule



dsp^2 hybridization - trigonal bipyramidal shape of $Fe(CO)_5$ molecule

94	20000	200	10
40	25000	150	2000
54.000	40000	30000	
20.000	85		
34.000			

7) Calculate the CFSE (Crystal Field Stabilization Energy) for each of the following system:

(a) d^1 (high spin octahedral)

For high spin d^1 octahedral complex the splitting is given by

$$CFSE = [-0.4 \times 3 + 0.6 \times 1] \Delta_0$$

$$= [-1.2 + 0.6] \Delta_0$$

$$= -0.6 \Delta_0$$

(b) d^6 (low spin octahedral)

low spin complex of d^6 having $\Delta_0 > P.E$

configuration is $t_{2g}^6 e_g^0$ and 3 electrons are paired in t_{2g} orbital.

$$= \frac{-2}{5} \Delta_0 \times 6 + 3P$$

$$= -\frac{12}{5} \Delta_0 + 3P$$

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Q#4: $Ni(CO)_4$ is diamagnetic and has tetrahedral structure. Explain.

Ans: The presence of σ ligand, which is a strong ligand, can pair all electrons in $Ni(CO)_4$ and thus its diamagnetic in nature. Tetrahedral shape of $Ni(CO)_4$ results from sp^3 hybridisation of Ni-atom.

Q5: Diff: b/w p-type and n-type semiconductor.

Ans:-

N-type

P-type

$\Rightarrow$ It is an intrinsic semiconductor which is obtained by doping the pentavalent atoms impurity atoms such as antimony, arsenic etc. to pure germanium or silicon semiconductor.

$\Rightarrow$ The impurity atoms added, provide extra electrons in the $\Rightarrow$ The impurity atoms added, create vacancies

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Structure and are of electrons (ie holes) called donor atoms. in the structure and are called acceptor atoms

⇒ The electrons are majority charge carriers and holes are minority charge carriers

⇒ The roles

are majority charge carriers and minority charge carriers

Q#6: O_2 is paramagnetic while O_2^{2-} is diamagnetic. Explain on basis of MOT.

Ans: To form O_2^{2-} , an extra electron enters the $2p$ orbital and one more electron enters the $2p$ orbital. Hence, the two orbitals are empty and all other orbitals have paired electrons. Thus this molecule is diamagnetic whereas, according to MOT there is 1 ~~unpaired~~ unpaired electron in $\pi 2p$ antibonding orbital and another unpaired electron in $\pi 2p$ antibonding orbital.

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Q7: Discuss factors influencing Δ_o .

Ans: Factors effecting Δ_o :-

- The arrangement of ligands around the metal ion.
- The coordination no. of the metal (i.e. tetrahedral, octahedral).
- The nature of the metal ion.
- The metal's oxidation state.
- The nature of ligands surrounding the metal ions.

"Past Paper - 2017"

Q1: Diff. b/w conductors, semiconductors and insulators on the basis of Band theory?

Ans: Conductors: Acc. to band theory, a conductor is simply a material that has its valence and conduction band overlapping, allowing electrons to flow through the material with minimal applied voltage.

Insulators: Insulator is a material which has a large forbidden gap b/w the energies of valence

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electrons and conduction band.

Semiconductors: Semiconductors have a small energy gap b/w the valence band and conduction band. Electrons can jump to conduction band, but not with the same ease as they do in conductors.

Q#2: Why CO is known as π -acceptor ligand?

Ans: CO as π -acceptor ligands:-

CO is a good π -acceptor (Lewis acid) due to empty π -orbitals and a good σ -donor (Lewis acid). When bonding to a metal the ligand (in this case CO) σ -donates to an empty d-orbital and the filled d-orbitals of metal donate to the empty π^* orbitals of CO, back donation.

Q#3: Give two limitations of VBT.

Ans: Limitations of VBT:-

→ Failure to explain tetrahedral exhibited by carbon.

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- No insight offered on the energies of electrons.
- The theory assumes that electrons are localized in specific areas.

Q#4: What are Chelates?

A: Chelates:

A chelate is an organic compound formed when a polydentate ligand bonds to a central metal atom. Chelation, according to IUPAC, involves the formation of two or more separate coordinate bonds b/w the ligand and central atom.

Q#5: What are inner orbital complexes? Give examples.

Ans. Inner orbital complex:-

inner

orbital complexes are coordination compounds composed of a central metal atom having hybridization of atomic orbitals including d-orbitals of inner shell and s, p orbitals from the outer shell.

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Q#16: Sodium is a good conductor while silicon is a semiconductor. why?

Ans: Reason:-

Sodium is a good conductor because it is a metal. On the other hand, silicon is a metalloid doped with impurities which make it a semiconductor.

"Past paper - 2016"

Q#1: What is the best achievement of CFT?

Ans: Achievements of CFT:-

CFT is a

bonding model that explains many imp. properties of transition-metal complexes including their colors, magnetism, structures, stability and reactivity. The central assumption of CFT is that metal-ligand interactions are purely electrostatic in nature.

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Q2. Why outer elements and d-block elements show variable oxidation states. why.

Ans. Reason: These variable elements show variable oxidation

state because their valence electrons in two diff. set of orbitals, that is $(n-1)d$ and ns .

The energy difference b/w these orbitals is ~~very~~ very less, so both the energy levels can be used for bond formation.

Q3. Describe soft x-ray spectra?

Ans. Soft x-ray spectra:-

Soft x-ray emission spectroscopy or (SXES) is an experimental technique for determining the electronic structure of materials. With a wavelength range b/w 200 eV - 3 keV, soft x-ray microscopes are capable of achieving spatial resolution of several hundred Angstroms ($\text{\AA}$).

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Q#4: Define and explain spectrochemical series?

Ans: Spectrochemical series:-

A spectrochemical series is a list of ligands ordered on ligand strength and a list of metal ions based on oxidation number, group and its identity.

"Past Papers - 2015"

Q#1: BF_3 molecule is planar but NH_3 is pyramidal. why?

Ans: Reason:-

BF_3 has planar shape while NH_3 has pyramidal shape because B has 3 electrons in outermost shell while N has 5 electrons in its outermost shell. Hence in NH_3 1 lone pair of electron is present and due to repulsion b/w lone pair and bond pair its shape become pyramidal.

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Q.2: Show hybridization of $\text{Cr}(\text{CO})_6$ with structure.

Ans: Structure: The electron diffraction study of $\text{Cr}(\text{CO})_6$ molecule in vapour state has shown that the molecule has octahedral shape.

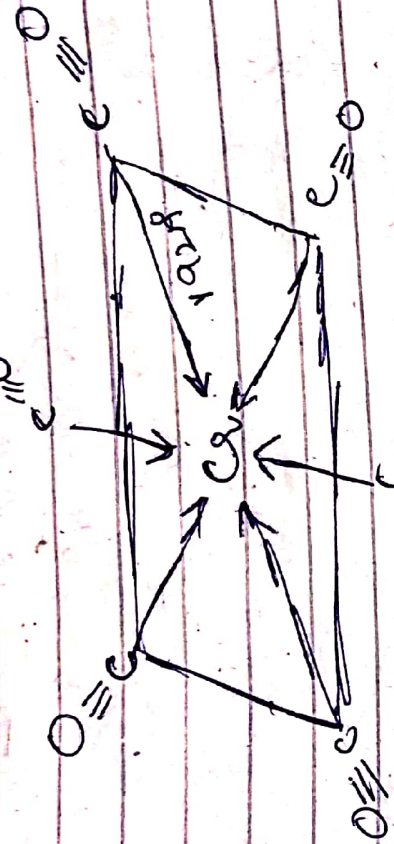
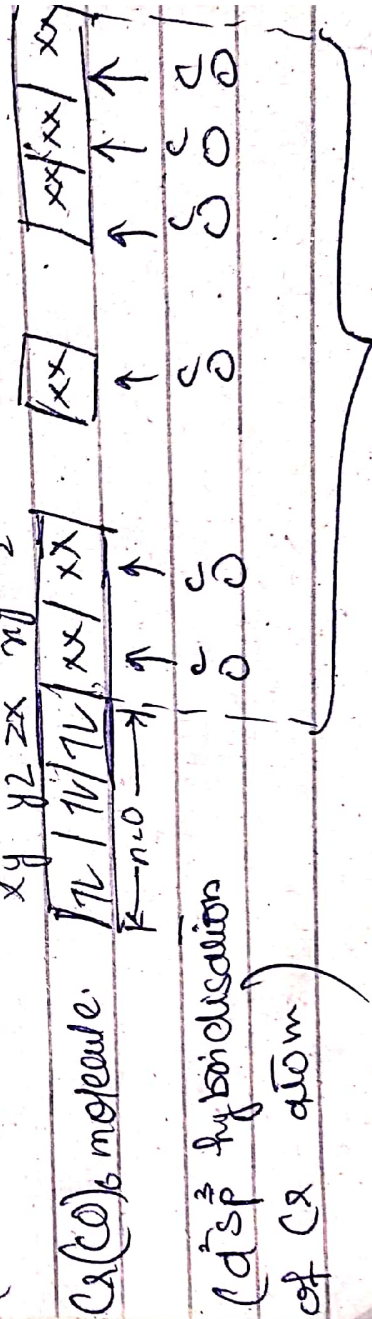
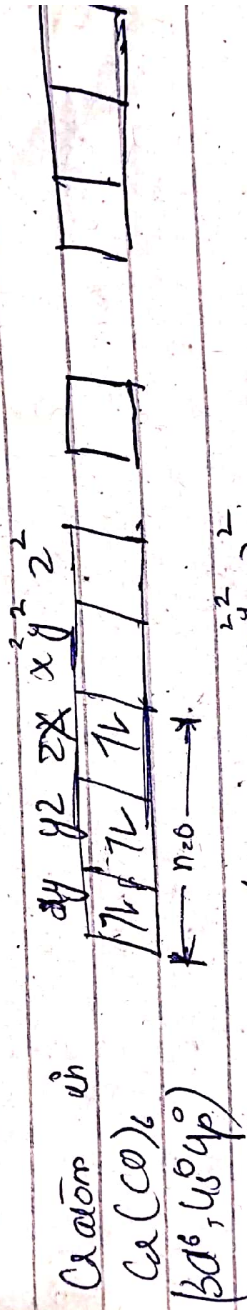
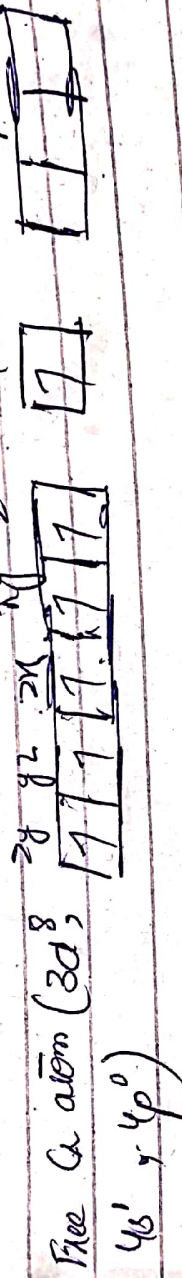


Fig 12.8. Octahedral structure of $\text{Cr}(\text{CO})_6$ molecule.

Hybridisation:- $\text{Cr} \rightarrow 3d \rightarrow 4s \rightarrow 4p$



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Q#3: write down two chemical methods

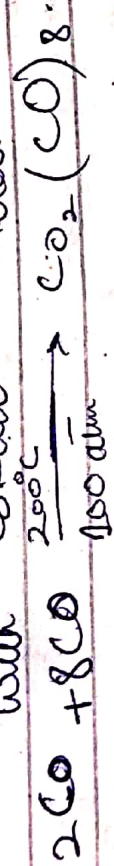
for the preparation of $\text{Co}(\text{CO})_8$.

Ans: Methods of Preparation:-

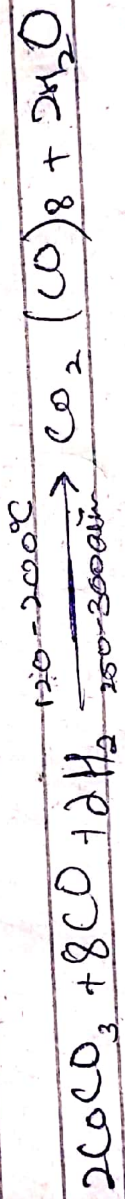
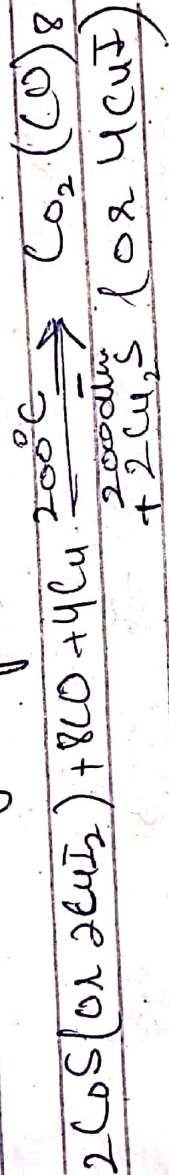
It is

prepared by:-

i) by the direct combination of CO with cobalt metal.



ii) by carbonylating CoS or CoI_2 with CO in presence of Cu-metal or by carbonylating CoCO_3 with CO in presence of H_2 . Here Cu-metal and H_2 both act as reducing agent.



~~Q#4: what is difference between~~

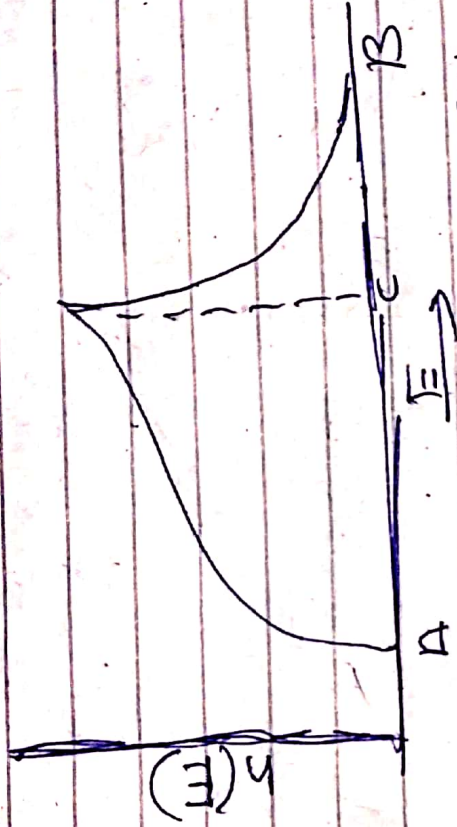
$n(E)$ and $N(E)$ waves?

Ans: $n(E)$ waves:-

$n(E)$ waves are used to show the complete distribution of electrons between the range of energy levels.

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in an energy band, $n(E)dE$ being the number of electrons per unit volume of material with energies b/w E and $(E+dE)$.



$n(E)$ plotted against E gives energy distribution of electrons in a band.

$N(E)$ curves :-

The $N(E)$ curves refer to the existing distribution of electrons b/w the various possible energy levels or states in a band.



Fig. Two $N(E)$ curves with an energy gap.